MILITARY SPECIFICATiON

HEAD ASSEMBLY: 10549216
(PERISCOPE: M20)

## 1. SCOPE

1.1 This specification covers a modular unit of the Tank Gunner's M20 Periscopes, herein referred to as “head assembly”.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-O-13810 | optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of |
| MIL-F-13926 | Fire Control Material; General Specification Governing the Manufacture and Inspection of |
| MIL-P-14232 | Parts, Equipment and Tools for Army Material, Packaging and Packing of |
| MIL-I-45607 | Inspection Equipment, Acquisition, Maintenance and Disposition of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |

DRAWINGS



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US Army, Frankford Arsenal

| | |
|---|---|
| 07560124 | Milli-Dioptometer, 20X |

DRAWINGS (Cont'd)

| | |
|---|---|
| F7560583 | Adapter |
| F7680994 | Fixture,SnapTest |
| F8565556 | PressureTester, Two-Station |
| FI0549216 | Head Assembly |
| F11738285 | Gage, Maximum Interchangeability |
| F11738286 | Gage, Minimum Interchangeability |

PACKAGING DATA SHEET

| | |
|---|---|
| 10549216 | Packaging of Head Assembly, 10549216 |

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Fabrication. The head assembly shall be manufactured in accordance with Drawing F10549216 and drawings pertaining thereto.

3.2 Interchangeability. The head assembly shall be capable of being secured onto the gages specified in 4.6.13 without restriction.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of Precedence
b. Dimensions and tolerances
c. inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.



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3.4.1 Storage temperature. The head assembly shall show no evidence of physical failure such as glass fracture or breakage, bonding or casting failure due to storage at temperatures ranging from -80° to + 160°F.

3.4.2 Operating temperatures. The head assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at ambient temperatures of -40° and + 150°F.

3.5 Torque. The running torque required to rotate the coupling through maximum elevation and depression shall not exceed 3 pound inches at a standard ambient temperature of 60° to 90°F. When checked at the temperature conditions specified in 3.4.2 the running torque shall not exceed 5 pound inches.

3.6 Orientation. The requirements of 3.7 and 3.9 shall be met with the head assembly posltioned as follows:

a. Mounting surface Datum Plane "B" (Ref. Dwg. F10549216 sheet 2) vertical.
b. Keyway Datum Line "A" (Ref.Dwg.F10549216 sheet 2) horizontal.
c. Coupling Datum Plane '*C" (Ref. Dwg. C8229213) vertical.

3.7 Collimation. The requirements of 3.7.1 to 3.7.6 inclusive shall be met with:

a. An incident line of sight which is originally perpendicular to Datum Plane B (see 3.6).
b. The input coupling positioned in accordance with 3.6.

3.7.1 Pre-vibration, Six Power (6X) System. The emerging line of sight through Window 8261623 shall be perpendicular to Datum Line A and Paralïel to Datum Plane B, both within 1 mil.

3.7.2 Unity Power System. The emerging line of sight through Window 8261620 shall be perpendicular to Datum Line A and parallel to Datum Plane B, both within 15 minutes.

3.7.3 Excursion range. Rotation of the input coupling shall cause the line of sight to deflect a minimum of 22 degrees above to 18 degrees below the horizontal plane.

3.7.3.1 Coukpling travel. Rotation of the input coupling shall be limited by contact with the coupling stop pin only.



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3.7.3.2 Repeatability . The coupling when snapped against pin on test fixture F7680994 in both directions, shall not result in a change in collimation in excess of 0.2 mil.

3.7.4 Plumb travel. Starting with the line of sight in coincidence with a vertical target Iine at the horlzontal position, the line of sight shall track the target line wit in 0.25 mil when elevated and depressed through the range specified in 3.7.3.

3.7.5 Elevation accuracy. This requirement shall be met with backlash removed. The change in angle at the input coupling shall cause the same angular change in the line of sight within 0.25 mil through the range specified in 3.7.3.

3.7.6 Backlash. Backlash between the line of sight and the input coupling shall not exceed 1 mil.

3.7.7 Resolution. Resolution thru Window 8261623 shall be 7 seconds of arc or less when measured with an observation telescope of at least 2.0 inch aperture and at least 20 power with the mirror set at 45° ± 2° relative to Datum Plane “B”. Resolution consists of detectable line structure and proper iine count in all four meridians, at one focus setting.

3.7.8 Sphericai Power & Astigmatism. The following requirements shall be met with the mirror set at 45° ± 2° relative to Datum Plane “B”.

3.7.8.1 Thru Window 8261623. Spherical power and astigmatism shall not exceed 4. millidiopters.

3.7.8.2 Thru Window 8261620. Spherical power and astigmatism shall not exceed 0.1 diopter when scanning the aperture with a dioptometer of at least 0.5 inch aperture.

3.8 Vibration. The Hlead Assembly shall show no evidence of physicai failure due to being in a vertical piane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/16 inch (1/8 total excursion) for a period of 5 minutes plus or minus 15 seconds.

3.9 Post-Vibration, 6X System. The emerging line of sight shall remain within 0.2 mil of the collimation reading established in 3.7.1.

3.10 Interior atmosphere. The Head Assembly shall have an interior atmosphere of nitrogen (dew point at ieast as low as -25° Fat atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig).



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3.11 Sealing. The Head Assembly shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5 ± 0.10 psig for a minimum of 2 hours.

3.12 Cleanliness.

a. Defects. There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end.
b. Dirt. There shall be no more than 3 particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.50.

## 4. QUALITY ASSURANCE PROVISlONS

4.1 Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) head assemblies plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.



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4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of Product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MlL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassembles shall be inspected in accordance with the inspection provisions contained m Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absents of SQAPS, the applicable Quality Assurance Provisions of Ml L-F- 13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MlL-STD-l05 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232, and Section 5 herein. The tabulated classification of defects in Tables I, II and Ill shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection. The following requirements and related tests shall be subjected to 100% Inspection:

| CHARACtERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| 1. Interchangeability | 3.2 | 4.6.13 |
| 2. Resolution | 3.7.7 | 4.6.6 |
| 3. Spherical power & astigmatism (Thru window 8261623) | 3.7.8.1 | 4.6.7.1 |
| 4. Spherical power & astigmatism (Thru window 8261620) | 3.7.8.2 | 4.6.7.2 |
| 5. Cleanliness | 3.12 | 4.6.12 |

NOTE: The tests of 4.3.2.2 shalt be conducted at a temperature between + 60° and + 90°F.



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TABLE I - CLASSIFICATION OF DEFECTS

| CLASS CHARACTERISTIC | REQUIREMENTS | TEST PROCEDURE |
|---|---|---|
| CRITICAL: NONE | | |
| MAJOR: AQL 0.65% Defective | | |
| 101. Torque | 3.5<br>3.7.1 | 4.6.3<br>4.6.4.1 |
| 102. Collimation, Pre-vibration six power (6X) system | 3.7.2 | 4.6.4.2 |
| 103. Collimation, unity power system | 3.7.3 | 4.6.4.3 |
| 104. Excursion range | 3.7.3.1 | 4.6.4.4 |
| 105. Coupling travel | 3.7.4 | 4.6.4.5 |
| 106. Plumb travel | 3.7.6 | 4.6.4.6 |
| 107. Backlash | 3.7.5 | 4.6.4.7 |
| 108. Elevation accuracy | 3.7.3.2 | 4.6.5 |
| 109. Repeatability | 3.8 | 4.6.8 |
| 110. vibration | 3.9 | 4.6.9 |
| 111. Collimation, post-vibration, six power (6X) system | 3.10 | 4.6.10 |
| 112. Interior atmosphere | 3.11 | 4.6.11 |
| 113. Sealing | | |
| MINOR: None | | |

NOTE: The tests of Table I shall be conducted at a temperature between + 60° and + 90°F.

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. One head assembly shall be selected at random as a special sampling from each produced. The sample shall meet the requirements and tests in Table II.



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TABLE II

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| 301. Fabrication | 3.1 | Applicable drawing - visual |
| 302. General specification | 3.3 | MIL-F-13926 - visual |

4.4.2 Environmental. Three (3) head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III, and shall then meet the requirements and tests in Table I.

TABLE III

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| 303. Storage temperatures (-80° and + 160°F) | 3.4.1 | 4.6.1 |
| 304. Low temperature operation (-40°F) | 3.4.2, 3.5 | 4.6.2, 4.6.3 |
| 305. High temperature operation ( + 150°F) | 3.4.2, 3.5 | 4.6.2, 4.6.3 |

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract all inspection equipment shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).



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4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing. The pressure testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance. Special testing equipment utilized to mount and orient the head assemble (see 3.6) for the performance tests shall conform to the following requirements:

a. The testing equipment shall be capable of securely mounting the head assembly with Datum Plane "B" (Ref. Dwg. F10549216) in a vertical plane within ten (10) seconds, Datum Line "A" (Ref. Dwg. FI0549216) in a horizontal plane within ten (10) seconds and solidly arresting Datum Plane "C" of the coupling (Ref.Dwg.C8229131) in a vertical position within ten (10) seconds. The test equipment shall also provide for articulation of Datum Plane "C" of the coupling to any angular position between its travel stop pin. The necessary Iocating and head mounting features of the Mount Periscope, Drawing F7652074 shall e incorporated into the testing equipment. Features shall also be designed into the testing equipment that will assure positive and accurate, positioning, and repositioning (repeated removal and replacement) of the head assembly on the testing equipment within five (5) seconds in all planes.

b. The testing equipment shall provide for positioning two target projector collimators, having a clear aperture of at least 2.0 inches, in the following manner: One target projector collimator shall be positioned at the entrance window of the head assembly so that its optical axis is in alinement with the line of sight emitting from the central aperture (geometric center) of the Head Window 8261623. The target reticle of the projector collimator shall have an



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appropriate calibrated horizontal and vertical reticle crossline target test pattern. The vertical reticle line of the target test pattern shall beat least 46 mils in length (23 mils on each side of the horizontal reticle line) to determine the excursion range specified in 3.7.3. The vertical reticle target line shall be adjusted plumb and the projector collimator shall beset at infinity focus. The other target projector collimator shall be positioned at the entrance window of the head assembly so that its optical axis is in alinement with the line of sight emitting from the central aperture (geometric center) of the Head Window 8261620. The target reticle of the projector collimator shall have an appropriate calibrated horizontal and vertical crossline target test pattern containing a 10 mil diameter circle at the intersecting point of the horizontal and vertical reticle lines to determine conformance with collimation (Unity power system), as specified in 3.7.3. The vertical reticle target line shall be adjusted plumb and the projector collimator shall beset at infinity focus. The two target projector collimators shall be autocollimated to the mounting surfaces provided on the test equuipment that positions Datum Plane “B” and Datum Line “A” (Ref. Dwg. F10549216 of the head assembly.

c. The testing equipment shall also provide for positioning two penta prisms having a deviation of ±5 seconds (calibrated), so that the optical axis of one penta prism is in alinement with the line of sight emitting from the central aperture (geometric center) of the Head Window 8261623, and the optical axis of the other penta prism is in alinement with the line of sight emitting from the central aperture (geometric center) of the Head Window 8261620.

d. Two angle reading telescopes, having a clear aperture of at least 2.0 inches and the capability of reading line of sight deviations, in the “X” and “Y” coordinates in seconds of arc, shall be a part of the testing equipment. The testing equipment shall provide for positioning the two an gIe reading telescopes at the penta prisms so that the optical axis of one angle reading telescope is in alinement with the line of sight emitting from the central aperture (geometric center) of the Head Window 8261623 and the optical axis of the other angle reading telescope is in alinement with the line of sight emitting from the central aperture (geometric center) of the Head Window 8261620. The optical axis of each of the two angle reading telescopes shall be autocollimated to the mounting surfaces provided on the test equipment that positions Datum Plane “B” and Datum Line “Y” (Ref. Dwg. F1O5492 16) of the head assembly. The horizontal and vertical reticle lines of the angle reading telescopes shall be superimposed on the target image lines prior to positioning the head assembly in the line of sight. The angle reading telescopes shall be adjusted for infinity focus.

e. Autocollimation and alinement shall be accomplished with the head assembly



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removed from the line of sight through the target projector collimators, penta prisms and angle reading telescopes. A setup device containing a front surface reflecting mirror shall be utilized for the alinement of the target projector collimators, penta prisms and angle reading telescopes. The necessary locating and mounting features of the head assembly shall be incorporated into the setup device.

f. All setup and calibration equipment with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and become a part of the testing equipment.

4.5.2.3.3 Repeatability (coupling). Special testing equipment to test collimation repeatability of the coupling (when snapped) shall conform to the requirements of Drawing F7680994, Fixture, Snap Test.

4.6 Test methods and procedures.

NOTE 1: Unless otherwise specified, caution shall be exercised during the following tests to prevent the input coupling from striking hard against its travel stop pin.

4.6.1 Storage temperature. The head assembly subjected to this test shall be pressureized with dry nitro en to an internal pressure between 0.25 and 0.50 pounds per square inch gage (psig). The nitrogen gas shall have a dew point at least as low as -25°F. The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note 2) to -80°F. The head assembly shall remain at this temperature for a minimum of 2 hours. At the completion of this 2 hour period, the temperature of the test chamber shall be gradually raised to + 160°F. The head assembly shall remain at this temperature for a minimum of 2 hours. At the completion of this 2 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the head assembly removed from the test chamber. The head assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.8. The head assembly shall conform to the requirements of 3.4.1.

NOTE 2: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures. The head assembly subjected to this test shall be pressurized with dry nitrogen to an internal pressure between 0.25 and 0.50 psig. The nitrogen gas shall have a dew point at least as low as -25°F. The testing



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equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MlL-F-13926 and the conditions of 3.4.2. The head assembly shall be" placed in the test chamber and the temperature of the chamber reduced gradually (see Note 2) to -40°F. The head assembly shall remain at this temperature for a minimum of 2 hours. At the completion of this 2 hour period, and while at -40°F, the head assembly shall be subjected to the test in 4.6.3. The head assembly shall conform to the applicable requirement of 3.5 for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised to +150°F. The head assembly shall remain at this temperature for a period of 2 hours. At the completion of this 2 hour period and while at +150°F, the head assembly shall again be subjected to the test in 4.6.3. The head assembly shall conform to the applicable requirement of 3.5 for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) and the head assembly removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the head assembly to thermal shock.

4.6.3 Torque. Standard torque measuring equipment capable of measuring the d torque values specified in 3.5 shall be utilize for this test. An appropriate adapter to accommodate the coupling shall be used with the torque measuring equipment. The running torque required to rotate the coupling through its full range of travel in each direction, shall conform to the applicable requirements of 3.5 for the temperatures indicated.

4.6.4 Orientation. The foilowing tests shall be performed with the head assembly positioned as specified in 3.6, utilizing special testing equipment that conforms to the requirements of 4.5.2.3.2. Prior to positioning the head assembly on the testing equipment, the horizontal and vertical reticle lines of the angle reading telescopes shall be superimposed on the horizontal and vertical target image lines. Unless otherwise specified herein, these tests shall be performed at the Head Window 8261623, and Datum Plane "C" of the coupling shall be in the vertiical position.

4.6.4.1 Collimation, pre-vibration, six power (6X) System. View through the angle reading telescope positioned at the Head Window 8261623 and observe the projected target image. The intersecting point of the horizontal and vertical reticle lines of the angle reading telescope shall be superimposed on the intersecting point of the horizontal and vertical target image lines within the tolerance specified in 3.7.1, as measured on the angle reading telescope. Record this collimation measurement for use in the subsequent test of 4.6.9. Upon completion of this collimation, pre-vibration test the head assembly shall then be subjected to the tests specified in 4.6.4.2 through 4.6.4.7 inclusive.



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4.6.4.2 Collimation, unity power system. This test shall be performed at the head Window 8261620. View through the angle reading telescope and observe the projected target image. The intersecting point of the telescopes horizontal and vertical reticle lines shall be within the circle on the target image to be in conformance with 3.7.2. Unless otherwise specified, subsequent testing shall be performed at the Head Window 8261623.

4.6.4.3 Excursion range. View through the angle reading telescope and if necessary, position the horizontal and vertical reticle lines of the telescope coincident with the horizontal and vertical target image lines. While observing the intersecting point of the telescope's horizontal and vertical reticle lines, rotate Datum Plane "C" of the input coupling so that the intersecting point of the horizontal and vertical target image lines is deflected from 22° above to 18° below the initial coincident position establish herein, as measured on the target image. Rotation of the input coupling shall conform to the requirements of 3.7.3.

4.6.4.4 Coupling travel. Rotate the input coupling to its maximum elevation, then to its maximum depression stop position. While the coupling is at each stop, back off the coupling for at least 1/4 rotation and then again return the coupling to the stop. Perform this a minimum of 3 times. During these rotations, the coupling shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metal click can be heard to verify that coupling rotation is being limited only by means of contact with the coupling stop pin and not by contact of other components (optical or mechanical) to be in compliance with 3.7.3.1.

4.6.4.5 Plumb travel. This test can be performed at the same time that the excursion range test (4.6.4.3) is performed. View through the angle reading telescope and establish the horizontal and vertical reticle lines of the telescope coincident with the horizontal and vertical target image fines. While observing the intersecting point of the telescope's horizontal and verticai reticle lines with respect to the vertical reticle tine of the target image, rotate Datum Plane "C" of the input coupling so that the intersecting point of the horizontal and vertical target image lines is deflected from 22° above to 18° below the coincident position established herein. During these rotations, the line of sight through the intersecting point of the telescope's horizontal and vertical reticle lines shall track the vertical target line within the tolerance specified in 3.7.4, as measured on the angle reading telescope.

4.6.4.6 Backlash. With backlash removed for direction of travel, position Datum Plane "C" input coup;ing vertical. View through the angle reading telescope and position the horizontal and vertical reticle lines of the telescope coincident with the horizontal and vertical target image lines. Rotate the coupling approximately



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10° in the same direction of travel then reverse rotation of the coupling (backlash included) until Datum Plane "C" of the coupling is repositioned vertical without overtravel. View through the angIe reading telescope and observe for coincidence of the telescope's horizontal reticle line with respect to the horizontal target image line. Any adjustment of the angle reading telescope to re-obtain coincidence of the telescope's horizontal reticle line with respect to the horizontal target image line is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.7.6.

NOTE: During this operation (4.6.4.6), if overtravel occurs, the test shall be nullified and the procedure specified for backlash shail again be performed.

4.6.4.7 Eievation accuracy. The testing equipment specified in 4.5.2.3.2 shall be utilized for this test except the target projector collimator positioned at the entrance window of the head assembly (in line with the Head Window 8261620), is not required. Auxiliary testing equipment empioying an auto-collimation device, affixed to the input coupiing of the head assembly shall be used in place of the target projector collimator. This testing equipment consists of an arm assembly with a calibrated optically fiat front surface mirror and an appropriate adapter that accommodates the input coupling of the head assembly. The mirror shall be auto-collimated to the angle reading telescope, at the Head Window 8261620, when Datum Plane 'C" of the input coupling is positioned vertical. The reflecting (auto-collimating) surface of the mirror shall lay in a plane perpendicular to Datum Plane "B" and parallel to a vertical center line passing through the geometric axis of the shaft of the input coupling adapter. The reflecting (auto-collimating) surface of the mirror shall also remain parallel to the vertical center line of the input coupling adapter during rotation through the range specified in 3.7.3. Affix and orient the auxiliary testing equipment on the input coupling of the head assembiy. With backlash removed for travel in elevation, position Datum Plane "C" of the input coupling vertical. Direct a line sight through the angIe reading teiescope and note the position of the reflected image with respect to the real image. Rotate Datum Plane "C" of the input coupling so that the line of sight is deflected to at least 22 degrees above the horizontal plane, then reversing direction of rotation, deflect the iine of sight to at least 18 degrees below the horizontal plane. During this rotation of the input coupling the reflected reticle image shall not deviate from the real reticle image beyond the tolerance specified in 3.7.5, as measured on the angle reading telescope. The head assembly shall then be subjected to the test in 4.6.8 (vibration).

4.6.5 Repeatability. The test fixture specified in 4.5.2.3.3 shall be utilized for this test. The test shall be performed in accordance with the instructions specified on the test fixture drawing. Any change in collimation, when the coupling face is snapped



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against pin on the test fixture in both directions, shall not exceed the tolerance in 3.7.3.2.

4.6.6 Resolution. Position the head assembly on its side with the mirror immobilized at 45° ± 2° relative to Datum Plane "B". The test for resolution shall be performed utilizing an observation telescope conforming to the requirements of 3.7.7, a resolving power chart on a wall target representing the angular subtense for the seconds of are specified for resolution, and a supporting medium to receive this test equipment as well as the item to be tested. The test method and procedure shall be in compliance with the "Resolution Test" specified in MIL-O-l3830, and the resolving power chart shall contain four line sets as shown in Figure 2 therein. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the revolving power chart. Resolution shall be within the value specified in 3.7.7, as determined by viewing through the observation telescope while scanning the clear exit aperture of the Head Window 8261623.

4.6.7 Spherical Power & Astigmatism. The following tests (4.6.7.1 & 4.6.7.2) shall be performed with the mirror immobilized at 45° ± 2° relative to Datum Plane "B".

4.6.7.1 Thru Window 8261623. With the head assembly positioned as in 4.6.6 and utilizing the resolving power chart specified therein together with a milli-dioptometer, astigmatism and spherical power shall be checked in accordance with the set-up and operating instructions on Dwg. D7560124. The milli-dioptometer shall conform to Dwg. D7560124 and shall be positioned so that its optical axis is perpendicular to the exit aperture of the Window, 8261623 of the Head assembly. Astigmatism and spherical power of the head assembly through the Window, 8261623 shall not exceed the tolerance (milli-diopters) specified in 3.7.8.1, as measured on the miili-diopter scale.

4.6.7.2 Thru Window 8261620. Perform this test in accordance with 4.6.7.1 except, the mitli-dioptometer shall be positioned so that its optical axis is perpendicular to the exit aperture of the Window, 8261620 shall be scanned. Astigmatism and spherical power of the head assembly through the Window, 8261620 shall not exceed the tolerance specified in 3.7.8.2 at any point in the scanning area as measured on the milli-diopter scale.

4.6.8 Vibration. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.8 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with "Test Facilities" requirements of MIL-F-13926. The head assembly shall be clamped on the vibrating equipment by means of an adapter that conforms to Drawing F7560583, Vibration



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Adapter for Head Assembly. The vibration motion shall be in the vertical plane for the time duration specified in 3.8. During this test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stop pin. Immediately following vibration, and prior to the examination or physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the test in 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the test in 4.6.9, the head assembly shall be thoroughly examined for physical failure resulting from poor workmanship and materials to determine compliance with 3.8. A visual examination shall be made for missing or damaged parts, glass chips, foreign matter and bonding (separations, chipping or rupture). A tactile examinations all be made for functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the head assemby for loose internal optical and mechanical components. Any one of the possible effects defined above shall be cause to consider a head assembly defective. Head assemblies shall have passed the above examinations to be in compliance with 3.8.

4.6.9 Collimation, post-vibration, six power (6X) system. Perform this test in accordance with 4.6.4.1. Record the collimation measurement. Any difference from the recorded collimation measurement established in 4.6.4.1 shall not exceed the tolerance specified in 3.9, as determined by the recorded values.

4.6.10 Interior atmosphere. This test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the head assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the head assembly by means of the dew point tester. Interior atmosphere of the head assembly shall conform to the requirements of 3.10.

4.6.11 Sealing. Internal pressurizing of the head assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the head assembly with dry nitrogen gas, having a dew point at least as low as minus 25°F, to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing on Drawing F8565556. Retain the applied pressure in the head assembly for a minimum of 2 hours. During the first hour, the input elevation shaft shall be rotated through a minimum of 3 full excursions of its rotational ran e. At the end of the 2 hour period there shall be no evidence of leakage in excess of 0.2 psig as indicated on the pressure gage of the testing equipment, to determine compliance with 3.11. Upon satisfactory completion of the sealing test, the head assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the head



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assembly shall be periodically sampled through a dew point tester until the atmosphere and pressure conforms to 4.6.10.

4.6.12 Cleanliness. The optical surfaces of the head assembly shall be inspected in accordance with MIL-O-13830. The head assembly shall conform to the requirements of 3.12.

4.6.13 Interchangeability. This test shall be performed utilizing the maximum and minimum Interchangeability gages in Drawings F11738285 and F11738286 respectively, to determine compliance with 3.2.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10549216. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 Intended use. The Head Assembly, 10549216 is a modular unit of the Tank Gunner's Periscope, M20 Series primarily used in the fire control system of the 90mm Gun Tank M47 and M48.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Applicable packaging data (see 5.1)
c. Selection of applicable levels of preservation, packaging and packing
d. Applicable stock number
e. Provisions for first article testing

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Project NO. 1240-A301



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

**INSTRUCTIONS:** This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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